Fact Strategies: Turnarounds and Break-Aparts
Answer Key
Grade 3 Math

Quick Answers

1. 12	4. 56, 56	7. 54, 54	10. 7
2. 16	5. 45, 45	8. 72, 72	
3. 42, 42	6. 4	9. 63, 63	

Curriculum

Level: Grade 3 Math
3.OA.A.3 – *problem 10*
3.OA.C.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9*
Difficulty 1–4 of 5 across 16 tagged checks.

1. Ana already knows that $4 \times 3 = 12$. Use the turnaround fact to find 3×4.
Solution: Swapping the two factors does not change the product — four rows of three counters and three rows of four counters hold the same number. So 3×4 has the same answer as 4×3.
$\Rightarrow$ $\boxed{12}$

2. Ben already knows that $8 \times 2 = 16$. Use the turnaround fact to find 2×8.
Solution: Same turnaround: 2×8 and 8×2 are the same fact written the other way round.
$\Rightarrow$ $\boxed{16}$

3. Break apart to multiply: $6 \times 7 = 6 \times (5 + 2)$
Solution: The 7 has been split into $5 + 2$, so multiply 6 by each piece and add:

$$6 \times 5 = 30, \qquad 6 \times 2 = 12, \qquad 30 + 12 = 42$$

$\Rightarrow$ $\boxed{42}$

4. Break apart to multiply: $7 \times 8 = (7 \times 4) + (7 \times 4)$
Solution: The 8 has been split into two fours, so work out one half and double it:

$$7 \times 4 = 28, \qquad 28 + 28 = 56$$

$\Rightarrow$ $\boxed{56}$

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5. Find both products: 9×5 and 5×9

Solution: Both are the same fact turned around, so both give the same product. Counting by fives nine times, or by nines five times, lands on the same number.

$\Rightarrow$ $\boxed{45}$

Noticing this halves how many facts have to be memorised: learn one of the pair and the other comes free.

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6. Use a missing-factor fact to divide: $24 \div 6$

Solution: Dividing asks how many groups of 6 make 24. Think through the sixes: $6, 12, 18, 24$ — that is four of them, so $6 \times 4 = 24$.

$\Rightarrow$ $\boxed{n = 4}$

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7. Nines are easier from tens: $9 \times 6 = (10 \times 6) - (1 \times 6)$

Solution: Nine groups of six is one group short of ten groups of six:

$$10 \times 6 = 60, \qquad 1 \times 6 = 6, \qquad 60 - 6 = 54$$

$\Rightarrow$ $\boxed{54}$

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8. Break 8×9 apart into two easier facts, then write the product.

Solution: Any split of one factor works. Splitting the 9 into $5 + 4$ gives two facts most students already know:

$$8 \times 5 = 40, \qquad 8 \times 4 = 32, \qquad 40 + 32 = 72$$

Other splits reach the same product — for example $8 \times 9 = (8 \times 10) - 8$. Accept any correct pair of facts with the right total.

$\Rightarrow$ $\boxed{72}$

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9. Malik says 7×9 gives the same answer as 7×10 with one group of 7 taken away. Use his strategy.

Solution: Malik is right. Nine groups of seven is ten groups of seven with one whole group removed:

$$7 \times 10 = 70, \qquad 70 - 7 = 63$$

The number taken away is a whole group of 7, not 1.

$\Rightarrow$ $\boxed{63}$

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10. There are 35 stickers shared equally among 5 friends. Write the missing-factor fact, then find how many stickers each friend gets.

Solution: Sharing 35 into 5 equal groups asks: five times what makes thirty-five? The matching fact is $5 \times n = 35$, and the fives run $5, 10, 15, 20, 25, 30, 35$ — seven of them. Each friend gets seven stickers.

$\Rightarrow$ $\boxed{n = 7}$